

TRETA SYSTEM

Comprehensive System Architecture, Data Lifecycle & Compound Delay Engine Manual

System Scope: Core Railway Simulation Engine, Dual-Database Sandbox, 5-Stage Physical Model & Dynamic RailRadar Telemetry
Generated Date: September 13, 2026 • 12:24 IST | **Status:** Production Verified (100% Endpoints Operational)

1. Executive Summary & Full Endpoint Operational Audit

The **TRETA SYSTEM** is an enterprise-grade railway simulation and compound predictive ETA platform. It combines real-time physical kinetic modeling, network constraint dispatching, and dynamic buffer slack absorption with strict dual-database isolation. All 15 system API endpoints were audited against the active backend server (<http://localhost:8000>) and confirmed 100% operational with HTTP 200 responses.

#	Method	Route / Endpoint	Operational Purpose & Parameters	Status	Payload Verification
1	GET	/api/simulation/status	Database sandbox status check	200 OK	mode: PRISTINE, is_pristine: true
2	GET	/api/search?q=12001	Fast train search/autocomplete	200 OK	Matched NDLS HBJ SHATABDI (12001)
3	GET	/api/train_info?train_no=12001	Full timetable metadata & hops	200 OK	Tier 1, route dist, EA slack, stops
4	GET	/api/corridors	7 National Corridors	200 OK	DEL-MUM, DEL-HWH, DEL-MAS, etc.
5	GET	/api/state_borders	29 State Border divisions	200 OK	29 border keys (SB-DL, SB-MH, etc.)
6	GET	/api/segments?corridor_slug=DEL-MUM&limit=5	Corridor route segments	200 OK	5 sequential Treta segment hops
7	GET	/api/predict?train_no=12919&speedup;=10&sec;=15	5-stage physics compound engine	200 OK	Gross/Net delay, MPS speedup, ETAs
8	GET	/api/predict (multi_station_injections)	Per-station circumstance shocks	200 OK	Aggregates station-wise weather/TSR
9	GET	/api/railradar/live?train_no=12919	Real-time GPS tracking telemetry	200 OK	Live speed, delay, progress fraction
10	GET	/api/railradar/config	RailRadar credentials config	200 OK	API key status, preview, sample train
11	GET	/api/railradar/match?train_no=12919	Scenario matrix nearest match (Query)	200 OK	Matches MasterPnC scenario in WIN.db
12	POST	/api/railradar/match	Scenario matrix nearest match (Body)	200 OK	Accepts telemetry JSON & matches
13	GET	/api/railradar/auto_fetch_and_freeze	One-click fetch, freeze & predict	200 OK	Locks controls to live telemetry
14	POST	/api/simulation/push	Write what-if edits to sandbox	200 OK	Writes to WIN_SIMULATION.db only
15	POST	/api/simulation/reset	Restore sandbox to pristine baseline	200 OK	Master untouched; rollback complete

2. Data Lifecycle: Ingestion, Preprocessing, Calculation, Reuse & Management

A. Dual-Database Isolation Architecture

To prevent accidental corruption of baseline railway timetables while providing dynamic what-if dispatch simulations, TRETA SYSTEM implements an isolated dual-database architecture:

WIN.db (Master Database)	WIN_SIMULATION.db (Sandbox Working Copy)
Access Mode: Strictly Read-Only (file:WIN.db?mode=ro). Master Data: 10,620 train metadata records, 38,880 precomputed scenario combinations (MasterPnC), 1,843 corridor segments, and 9,950 station state border mappings. Integrity Guarantee: No user action, API request, or automated script can write to this file.	Access Mode: Read-Write Working Sandbox. Dynamic Alterations: Tracks four simulation columns in RouteDivision: sim_proposed_delay, sim_new_eta, sim_status, and sim_updated_at. Live Telemetry Persistence: Records incoming GPS snapshots into RailRadarTelemetry and secondary impacts into SimulationCascadedEffects. Atomic Rollback: /api/simulation/reset overwrites sandbox with pristine master copy.

B. Pre-Warmed In-Memory Data Caching & Sub-Millisecond Reuse

Upon application boot, AppServer.py indexes and retains core railway network topology in memory to eliminate SQL latency during real-time delay inference:

- **TRAIN_MAP & EACalculation:** 10,620 train profiles cached into hash tables for O(1) retrieval of train tier, total distance, scheduled halts, and baseline Extra Allotted Time (EA).
- **STATION_TIMETABLE_INDEX & HOP_INDEX:** Inverts 200,126 dynamic Treta section hops across 31,546 station pairs. Hops are instantly queryable by station code pairs (from_code, to_code) without touching disk.
- **MASTER_PNC_CACHE:** Pre-loaded 38,880 scenario permutations for instant O(1) matching against incoming RailRadar live telemetry.
- **STATION_STATE_MAP:** Lookup index mapping 9,950 stations to 29 state border division codes (SB-DL, SB-MH, SB-UP, etc.) for automated border crossing identification.

3. The 5-Stage Physical Compound Delay Engine

The delay engine simulates how primary exogenous shocks trigger secondary network constraints, which are mitigated by driver speedup catch-up and timetable buffer slack (\$EA\$) to determine final destination delay and station-by-station ETAs.

STAGE 1: Primary Disturbance Shock Layer (PD)

$$PD = D_{inherited} + D_{weather} + D_{TSR} + D_{ACP} + D_{traction} + D_{section} + D_{multi_station}$$

- **Inherited Delay ($D_{inherited}$):** Live operational delay ingested directly from RailRadar GPS telemetry.
- **Weather Disturbances ($D_{weather}$):** Clear (0m), Fog (+15m), Heavy Rain (+25m), Cyclone / Severe Weather (+60m).
- **Temporary Speed Restrictions (D_{TSR}):** None (0m), TSR 60 (+5m), TSR 45 (+10m), TSR 30 (+20m), TSR 15 (+35m).
- **Alarm Chain Pulling (D_{ACP}):** 0 Events (0m), 1 Event (+10m brake reset & investigation), 2 Events (+25m).
- **Locomotive Traction Defects ($D_{traction}$):** Nominal (0m), Minor Traction Defect (+15m), Major Engine Failure (+45m).
- **Section & Multi-Station Injections:** Direct user-injected corridor section delay plus cumulative station circumstance penalties.

STAGE 2: Secondary Network Constraint Layer (CD)

$$CD = D_{headway} + D_{crossing} + D_{platform} + D_{crew}$$

- **Headway & Priority Congestion ($D_{headway}$):** Track Clear (0m), Medium Congestion (+12m), Heavy Express Congestion (+30m), Severe Gridlock (+55m).
- **Single-Track Crossing Conflicts ($D_{crossing}$):** No Conflict (0m), Minor Wait (+8m), Major Cross / Siding Hold (+22m).
- **Terminal Platform Starvation ($D_{platform}$):** Platform Available (0m), Platform Occupied (+12m), Terminal Gridlock (+30m).
- **Crew Duty Hours of Service (D_{crew}):** Duty Valid <8h (0m), Duty Approaching Limit (+10m), Duty Expired ≥8h (+40m mandatory crew change).

STAGE 3: Gross Delay (GD) & Maximum Permissible Speed (MPS) Speedup Recovery

$$GD = PD + CD \mid D_{speedup_recovered} = \min(GD, D_{speedup_allowance}) \mid GD_{eff} = GD - D_{speedup_recovered}$$

When trains travel across unconstrained green-signal sections, locomotive pilots can accelerate up to the section Maximum Permissible Speed (MPS). The system calculates throttle margin recovery allowance (presets: 0m, 5m, 10m, 15m, 20m) to offset gross delay before entering timetable buffer zones.

STAGE 4: Timetable Buffer Slack (EA) Absorption

$$Max_Absorbed = EA * Tier_Recovery_Rate \text{ (TRR)} \mid D_{absorbed} = \min(GD_{eff}, Max_Absorbed)$$

Every scheduled timetable incorporates Extra Allotted Time (EA in minutes) engineered into run profiles. The absorption efficiency depends on the train category tier:

- **Tier 1 Premium (Vande Bharat / Shatabdi / Rajdhani):** 35% recovery rate.
- **Tier 2 Superfast (Superfast / Mail / Express):** 22% recovery rate.
- **Tier 3 Ordinary / Freight (Passenger / Goods):** 12% recovery rate.

STAGE 5: Net Destination Delay (ND) & Dynamic ETA

$$ND = \max(0.0, GD_{eff} - D_{absorbed}) \mid Predicted_Arrival_Clock = Scheduled_Arrival_Clock + ND$$

- **Status Classification:** ON_TIME (ND <= 5.0m), SLIGHT_DELAY (5m < ND <= 30m), MODERATE_DELAY (30m < ND <= 60m), MAJOR_DELAY (ND > 60m).
- **Ahead Stations Dynamic ETAs:** Stop-by-stop progressive simulation calculates distance traversed, applies local circumstance penalties, and computes actual arrival and departure clock times with scheduled platform dwell durations.

Cascading Ripple Propagation & Absorption Invariance Rule

- **Active Cascading Ripple (ND > 0):** When the primary train arrives late, headway blocks delay direct trailing followers (Train B inherits ~70% ripple) and secondary downstream followers (Train C inherits ~45% ripple).
- **Absorption Invariance Rule (ND <= 0):** When speedup catch-up or buffer slack (EA) absorbs all disturbance delay (Net Delay = 0.0), downstream propagation is strictly deactivated (has_cascading_delay = False, overlaid_trains = []), preventing false alarm cascades.

4. RailRadar Live GPS Integration & Telemetry Freezing

1. **Live GPS Telemetry Ingestion:** Automatically contacts RailRadar API for real-time delay minutes, instantaneous speed (km/h), segment progress fraction (0.0 to 1.0), and halt station status.
2. **Permutation Scenario Matching:** Matches live delay to the nearest precomputed scenario in WIN.db (MasterPnC 38,880 combinations).
3. **Control Freezing:** Locks UI sliders in *Operational Disturbance Cases* (is_frozen: true) to preserve real-world ground truth while live telemetry is active.
4. **Offline / Built-in Engine Fallback:** If no network signal or API key is available, the engine smoothly falls back to built-in sample telemetry (e.g. Train 12919 Malwa SF Express) with clean baseline data.

5. Complete Project Feature Matrix

Component / Layer	Key Features & Capabilities
Unified Branding & Header	Global TRETA SYSTEM branding, real-time database mode indicator (PRISTINE vs MODIFIED), and active national corridor badge.
Corridor & Route Selector	Instant search over 10,620 trains with autocomplete; dropdown filters for 7 National Corridors, 29 State Border Divisions, and individual route segment hops.
Operational Disturbance Cases	Unified 4-group control suite: Physical Disturbances (Weather, TSR, Traction), Network Constraints (Headway, Crossings, Platforms), Operational Rules (ACP, Crew HOS), and Multi-Station Circumstances & Injected Delay Engine.
Multi-Station Circumstance Injector	Allows dispatchers to pick any halt station along the train route, assign specific weather/TSR shocks, and view active circumstance chips with one-click removal.
Ahead Stations ETA Board	Real-time stop-by-stop timetable showing scheduled vs predicted arrival/departure clock times, dwell times, and accumulated delay across all upcoming stations.
Cascaded Ripple Visualizer	Visualizes ripple propagation across the track block from Primary Incident Train A to Trailing Train B and Downstream Train C.
Database Sandbox Manager	Interactive 'Push to Simulation DB' (writes what-if mutations to WIN_SIMULATION.db) and 'Reset Simulation DB' (atomic file rollback to master baseline).

6. Deployment & Runtime Operations

- **Backend API Server:** Running on http://localhost:8000 via python AppServer.py.
- **Frontend Web Application:** Running on http://localhost:3000 via npm run dev (or npm start).
- **Environment Configuration:** PostgreSQL / NeonDB connection, RailRadar API key, and master database paths configured in Backend/.env.